

Surprisingly, the KitKat update on Nexus 4 and Nexus 7 does not come with most of the above UI updates, for example, status and system bars still have the original black background.

“To make sure that users always have easy, consistent access to system UI from a full-screen immersive mode, Android 4.4 supports a new gesture—in immersive mode, an edge swipe from the top or bottom of the screen now reveals the system UI,” said sources at Google.

Enhanced search and assistant features

The leftmost home screen is the new home of Google Now, though it is still available through a swipe-up gesture on the system bar. The standalone launcher is now replaced with Google’s search app running as the home screen. Users can interact with the device in Siri fashion to search anytime with voice interaction. Voice search is initiated by speaking out the words, “Okay Google” to the home screen. You can ask about the weather, theatre timings and for directions. Or, you can get the device to play a certain song, text a friend, or make a phone call.

Google has also repurposed its *Hangouts* application to feature SMS and MMS messages, as well as IM threads from Google contacts. Locations and photos can also be shared over Hangout. Google Hangouts has replaced the messaging app in KitKat.

It is worth mentioning the Moto X device here, which has got a special in-built processor for Google voice search to work from any screen, not only from the home screen.

Instagram inspired

With KitKat’s photo editor, you can apply predefined looks, alter the geometry, the saturation and apply filters. “The editor is pretty powerful, works on tablets and phones, handles full-size image processing, zooming, re-editing, image exports, user presets, etc. This new version also adds more powerful specialised tools (graduated filters, per-channel saturation controls, local adjustments, etc),” a Google post reported. Another great addition is that users can now export their edited images in the size they want. This comes with an adjustable slider.

Sensors

Android 4.4 tracks what you do and where you are using sensors. This is possible because Google has enabled hardware sensor batching, a feature that makes sensors far less power hungry. It collects and delivers sensor events in batches, rather than individually, as they are detected. It allows the device application processor to remain in low-power idle state until batches are delivered. Applications can request for batched events and control their frequency.

Performance

Theoretically, Android KitKat will run on handsets with just 512 MB of RAM. It has streamlined all the major components

and optimised the overall system for reduced memory usage. It has also introduced new APIs and tools for developers to analyse memory usage in their applications.

The touch screen is more responsive and users will experience an enhanced and faster multitasking environment with quicker loading apps.

Printer support

Android KitKat supports native printing on its platform. It provides a print manager that integrates various print services available in the system and serves requests sent from applications.

“The print manager provides shared services and a system UI for printing, giving users consistent control over printing from any app. The print manager also ensures the security of the content as it is passed across processes, from an app to a print service.”

Manufacturers are free to develop printer apps that plug into these APIs for system-wide print support. The only printer OEM currently on board is HP, but there is a Google Cloud Print plugin that will support just about every printer.

This feature in Android is a big plus from the enterprise point of view.

Payment support

Mobile phone carriers can no longer have any control over who has access to the NFC chip for payments. Previous implementations of NFC in Android used a ‘secure element’ that stored credit card information on the device. Access to the secure element used to be provisioned by the device vendor; before, this came from carriers. KitKat uses ‘host-based card emulation’ (HCE). This allows any app access to tap-and-pay, with no approval needed from the device vendor. Google Wallet can be installed and used on any KitKat NFC device.

Video capture support

Android KitKat incorporates the capability to capture real-time video along with the option to share that video directly from the device.

Smart caller ID

Android KitKat supports configuring the contacts list with any of the social networks like LinkedIn, Facebook, Twitter, Google Mail, etc. It synchronises all the contact details including profile pictures with the respective network. It further tries to search calls from any unknown numbers with Google Maps and if found, it will use this information for the caller ID function.

Payments

This section will incorporate NFC payments into the operating system. This means that users can access their carrier pages or accounts directly through Android, instead of using third party apps.